

P8131 Spring 2026 Homework #10

Due on May 4 at 11:59pm

1. Suppose we have the following data:
Group 1: 4, 12+, 15, 21+, 23;
Group 2: 2, 6+, 8+, 10, 19.
Test $H_0 : h_1(t) = h_2(t)$ for all t versus $H_1 : h_1(t) \neq h_2(t)$ for some t using the log-rank test.
2. Consider the “kidtran” dataset in the KMsurv package where the death times of kidney transplant patients are reported. The patients can be classified by race (1=white and 2=black) and gender (1=male and 2=female) into one of four groups. Compare the survival curves for different races in each sex group.
3. Consider the “larynx” data in the KMsurv package. Let
 $Z1 = 1$ if the patient is in stage II, 0 otherwise;
 $Z2 = 1$ if the patient is in stage III, 0 otherwise;
 $Z3 = 1$ if the patient is in stage IV, 0 otherwise;
 $Z4 =$ age of the patient (in years).
Fit a proportional hazards model using the covariates $Z1, Z2, Z3, Z4$ and the interaction term $Z1 \cdot Z4$. Explain your results. What is the relative risk of dying for a stage II patient of age 50 as compared to a stage I patient of the same age?